

Division Algorithm, Remainder Theorem, Factor Theorem

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)
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Learning Outcomes

At the end of this lecture, you should be able to:

- ☐ Understand the division algorithm and know what is the divisor, quotient, and remainder.
- ☐ Understand and apply the Remainder Theorem.
- ☐ Perform the long division for polynomials.
- ☐ Perform the synthetic division for polynomials.
- ☐ Understand and apply the Factor Theorem.

Division Algorithm

Let $f(x)$ and $g(x)$ be two polynomials with $g(x) \neq 0$. Then there exist a unique polynomials $q(x)$ and $r(x)$ such that $f(x) = q(x)g(x) + r(x)$ where $r(x) = 0$ or $\deg r(x) < \deg g(x)$.

Handwritten notes:
 $7 = 1(3) + 4$
 $7 = 2(3) + 1$ ← 7 divided by 3, quotient is 2 and remainder is 1

Here, $q(x)$ is called the **quotient** and $r(x)$ is called the **remainder** in the division of $f(x)$ by $g(x)$.

Handwritten note: $2 = 0(4) + 2$

If $f(x) = 0$ or $\deg f(x) < \deg g(x)$, then $q(x) = 0$ and $r(x) = f(x)$.

If $\deg f(x) \geq \deg g(x)$, then $\deg f(x) = \deg g(x) + \deg q(x)$.

Remainder Theorem

Remainder Theorem:

If $f(x)$ is a polynomial and c is a constant, then the remainder in the division of $f(x)$ by $x - c$ is $f(c)$.

$$f(x) = q(x)(x-c) + \underbrace{r(x)}_{=f(c)}$$

proof. By the division algorithm, there exists polynomials $q(x)$ and $r(x)$ such that

$$f(x) = q(x)(x-c) + r(x)$$

where $r(x) = 0$ or $\deg r(x) < \deg \underline{(x-c)} = 1$.

So, $\deg r(x) = 0$

Then, $r(x)$ must be a constant and $r(x) = b$ for some constant b . So now, we have $f(x) = q(x)(x-c) + b$

when we substitute $x=c$, $f(c) = q(x)(c-c) + b$

$$\Rightarrow b = f(c)$$

$$\text{So, } f(x) = q(x)(x-c) + \underline{\underline{f(c)}}$$

and $f(c)$ is the remainder in the division of $f(x)$ by $x-c$.

Remainder Theorem

When $f(x)$ is divided by $ax - b$ or $x - \frac{b}{a}$, where $a \neq 0$, the remainders are the same (however the quotient are different).

$$f(x) = q(x) \left(x - \frac{b}{a}\right) + r \quad \leftarrow r = f\left(\frac{b}{a}\right) \text{ for some polynomial } q(x) \text{ and constant } r.$$

$$\Rightarrow f(x) = \frac{1}{a}q(x)(a(x - \frac{b}{a})) + r$$

$$\Rightarrow f(x) = \left[\frac{1}{a}q(x)\right](ax - b) + r$$

Example:

What is the remainder when $x^3 + (2 + 3i)x^2 - x - i$ is divided by $2x + 1$?

$$\text{Let } f(x) = x^3 + (2 + 3i)x^2 - x - i.$$

Since $2x + 1 = 2(x + \frac{1}{2}) = 2[x - (-\frac{1}{2})]$, by the remainder theorem, the remainder is $f(-\frac{1}{2})$.

$$f(-\frac{1}{2}) = (-\frac{1}{2})^3 + (2 + 3i)(-\frac{1}{2})^2 - (-\frac{1}{2}) - i = \frac{1}{8}(7 - 2i)$$

Example

What is the remainder when $f(x) = \underline{3x^4 + 12x^2 - 5x + 36}$ is divided by $x^2 + 2x - 3$?

By the Division algorithm,

$$\begin{aligned} f(x) &= q(x) [x^2 + 2x - 3] + r(x), \quad \deg r(x) < \deg (x^2 + 2x - 3) \\ &= q(x) [(x+3)(x-1)] + r(x) \end{aligned} \quad = 2.$$

Let $r(x) = ax + b$ for some constant a, b .

$$f(x) = \underline{q(x)(x+3)(x-1)} + ax + b. \quad \leftarrow \text{Find } a, b$$

$$f(1) = 3(1)^4 + 12(1)^2 - 5(1) + 36 = a + b$$

$$\Rightarrow a + b = 46$$

$$f(-3) = 3(-3)^4 + 12(-3)^2 - 5(-3) + 36 = -3a + b \Rightarrow -3a + b = 402.$$

$$a+b = 46 \text{ --- (1)}$$

$$-3a+b = 402 \text{ --- (2)}$$

$$(2) - (1), -4a = 356$$

$$a = -89$$

$$\text{From (1), } -89 + b = 46$$

$$b = 135$$

Therefore the remainder $r(x) = -89x + 135$

Long Division

The Remainder Theorem can only help us to find the remainder. Now we will learn some other methods to help us finding both quotient and remainder.

The first method we learn in the course is called **long division**.

For each of the following polynomials $P(x)$ and $d(x)$, find the quotient and the remainder when $P(x)$ is divided by $d(x)$:

(a) $P(x) = 1 + 2x^3 - 3x^2$, $d(x) = x - 2$;

(b) $P(x) = x + 2x^4 - 3x^3 + ix^2 - 2$, $d(x) = x^2 - 3x + 2$.

$$7 \div 3$$

$$\begin{array}{r} 2 \leftarrow \text{quotient} \\ 3 \overline{) 7} \\ \underline{6} \\ 1 \leftarrow \text{remainder} \end{array}$$

Long Division

(a) $P(x) = 1 + 2x^3 - 3x^2$, $d(x) = x - 2$

$$= \underline{2x^3 - 3x^2 + 1}$$

$$\begin{array}{r}
 2x^2 + x + 2 \quad \leftarrow \text{quotient} \\
 x-2 \overline{) 2x^3 - 3x^2 + 0x + 1} \\
 \underline{(-) 2x^3 - 4x^2} \\
 x^2 + 0x + 1 \\
 \underline{(-) x^2 - 2x} \\
 2x + 1 \\
 \underline{(-) 2x - 4} \\
 5
 \end{array}$$

you can check
↓

Therefore,

$$P(x) = (x-2)(2x^2 + x + 2) + 5$$

So, quotient = $2x^2 + x + 2$
remainder = 5

5 ← remainder.

$$\begin{array}{r}
 151 \quad \leftarrow \text{quotient} \\
 5 \overline{) 756} \\
 \underline{(-) 5} \\
 25 \\
 \underline{(-) 25} \\
 6 \\
 \underline{(-) 5} \\
 1 \quad \leftarrow \text{remainder}
 \end{array}$$

$$\deg r(x) < \deg d(x)$$

Long Division

$$(b) P(x) = \overbrace{x + 2x^4 - 3x^3 + ix^2 - 2}^{= (2x^2 + 3x + (5+i))(x^2 - 3x + 2) - (12 + 2i)}, d(x) = x^2 - 3x + 2$$

$$2x^2 + 3x + (i+5)$$

$$(x^2 - 3x + 2) \overline{) 2x^4 - 3x^3 + ix^2 + x - 2}$$

$$(-) 2x^4 - 6x^3 + 4x^2$$

$$3x^3 + (i-4)x^2 + x - 2$$

$$(-) 3x^3 - 9x^2 + 6x$$

$$(i+5)x^2 - 5x - 2$$

$$(-) (i+5)x^2 - (3i+15)x + (5i+10)$$

$$\text{so quotient} = 2x^2 + 3x + (5+i) \quad (3i+10)x - 2i - 12$$

$\deg r(x) < \deg d(x)$

$$\text{remainder} = (10+3i)x - (12+2i)$$

Synthetic Division

The second method we learn in the course is called **synthetic division**.

This method could be used if we have a linear divisor.

Consider the polynomial

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \quad (a_n \neq 0)$$

of degree n and the divisor $x - b$.

We want to find a quotient $q(x)$ and the remainder r such that

$$P(x) = \underbrace{(x - b)q(x)} + r \quad \begin{array}{l} \text{deg } r < \text{deg } (x - b) \\ (r \text{ is a constant, } \text{deg } q(x) = n - 1) \end{array}$$

Let $q(x) = c_n x^{n-1} + c_{n-1} x^{n-2} + \cdots + c_2 x + c_1$ and $r = c_0$, so we wish to find c_i ($i = 0, 1, 2, \dots, n$).

Synthetic Division

$$\begin{aligned}P(x) &= a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \\&= (x-b) q(x) + r \quad \leftarrow \text{from division algorithm.} \\&= (x-b) (c_n x^{n-1} + c_{n-1} x^{n-2} + \dots + c_2 x + c_1) + c_0 \\&= c_n x^n + (c_{n-1} - b c_n) x^{n-1} + (c_{n-2} - b c_{n-1}) x^{n-2} \\&\quad + \dots + (c_1 - b c_2) x + (-b c_1) + c_0\end{aligned}$$

So by ^(equating) comparing the coefficients, we obtain

$$\begin{array}{lll}x^n : & a_n = c_n & \Rightarrow c_n = a_n \\x^{n-1} : & a_{n-1} = c_{n-1} - b c_n & \Rightarrow c_{n-1} = a_{n-1} + b c_n \\x^{n-2} : & a_{n-2} = c_{n-2} - b c_{n-1} & \Rightarrow c_{n-2} = a_{n-2} + b c_{n-1} \\& \vdots & \vdots \\x^1 : & a_1 = c_1 - b c_2 & \Rightarrow c_1 = a_1 + b c_2 \\x^0 : & a_0 = c_0 - b c_1 & \Rightarrow c_0 = a_0 + b c_1\end{array}$$

Synthetic Division

$$c_{n-i} = a_{n-i} + b c_{n-i+1}$$

So we can obtain c_i by constructing the following table:

	x^n	x^{n-1}	x^{n-2}	$\dots$	x^2	x	$x^0 = 1$
b	a_n	a_{n-1}	a_{n-2}	$\dots$	a_2	a_1	a_0
+		bc_n	bc_{n-1}	$\dots$	bc_3	bc_2	bc_1
$a_n =$	c_n	c_{n-1}	c_{n-2}	$\dots$	c_2	c_1	c_0

$$q(x) = c_n x^{n-1} + c_{n-1} x^{n-2} + \dots + c_2 x + c_1, \quad r = c_0.$$

Synthetic Division

Example:

For each of the following polynomials $P(x)$ and $d(x)$, find the quotient and the remainder when $P(x)$ is divided by $d(x)$:

- (a) $P(x) = x^4 - 2x^3 + 4x - 1$, $d(x) = x - 3$;
 (b) $P(x) = 2x^3 + 3x^2 - (5 + 2i)x - 9i$, $d(x) = x + (2 - i)$;
 (c) $P(x) = 10x^4 + 7x^3 + x^2$, $d(x) = 2x + 1$;
 (d) $P(x) = 3x^4 + 12x^2 - 5x + 36$, $d(x) = x^2 + 2x - 3$.
 (a) $P(x) = x^4 - 2x^3 + 4x - 1$, $d(x) = x - 3$ $b=3$.

$x - b$

So quotient:
 $x^3 + x^2 + 3x + 13$
 remainder: 38.

b	x^4	x^3	x^2	x^1	x^0
3	1	-2	0	4	-1

(+)

$\downarrow 1 \times 3$	3	$\downarrow 1 \times 3$	3	$\downarrow 3 \times 3$	9	$\downarrow 12 \times 3$	39
c_4	1	c_3	1	c_2	3	c_1	13
							38

$\leftarrow r_0$ / remainder

Synthetic Division

(b) $P(x) = 2x^3 + 3x^2 - (5 + 2i)x - 9i$, $d(x) = x + (2 - i)$

$$= x - (\underline{i-2}) \quad b = i-2$$

b	x^3	x^2	x^1	x^0
<u>$i-2$</u>	2	3	$-5-2i$	$-9i$

$$(2i-1)(i-2)$$

$$= 2i^2 - 4i - i + 2$$

$$= -2 - 5i + 2 = -5i$$

(+)

	$\downarrow$	$2i-4$	$-5i$	$17+9i$
$\swarrow$	<u>2</u>	$2i-1$	$-5-7i$	17

coefficients of quotient $\rightarrow$ remainder $\leftarrow$

So quotient = $2x^2 + (2i-1)x + (-5-7i)$

remainder : 17

Synthetic Division

(c) $P(x) = 10x^4 + 7x^3 + x^2$, $d(x) = \underline{2x + 1}$;

$$\begin{aligned} P(x) &= \underline{q(x)} \underline{d(x)} + \underline{r} \\ &= q(x) (\underline{2x+1}) + r \\ &= 2q(x) (\underline{x + \frac{1}{2}}) + r \\ &= \boxed{2q(x)} (\underline{x - (-\frac{1}{2})}) + r \end{aligned}$$

$$\begin{array}{r|rrrrr} & x^4 & x^3 & x^2 & x^1 & x^0 \\ -\frac{1}{2} & 10 & 7 & 1 & 0 & 0 \\ \hline (+) & & -5 & -1 & 0 & 0 \\ \hline & 10 & 2 & 0 & 0 & 0 \end{array}$$

coefficient of $2q(x)$

$$\text{So, } 2q(x) = 10x^3 + 2x^2 \Rightarrow q(x) = 5x^3 + x^2$$

$$\text{quotient: } 5x^3 + x^2$$

$$\text{remainder: } 0.$$

Synthetic Division

(d) $P(x) = 3x^4 + 12x^2 - 5x + 36$, $d(x) = x^2 + 2x - 3 = (x-1)(x+3)$

$$p(x) = \underbrace{q(x)}_{?} (x^2 + 2x - 3) + \underbrace{r(x)}_{?}$$

$$= q(x)(x-1)(x+3) + r(x)$$

$$= \boxed{q(x)(x+3)} \underbrace{(x-1)}_{x-1} + r(x)$$

you can
separate into
different
synthetic
divisions

$$q(x)(x+3)$$

	x^4	x^3	x^2	x^1	x^0
$1 \mid$	3	0	12	-5	36
$(+)$		3	3	15	10
$-3 \mid$	3	3	15	10	46
		-9	18	-9	
	3	-6	33	-89	

So $p(x) = (3x^3 + 3x^2 + 15x + 10)(x-1) + 46$.

$$= \boxed{(3x^2 - 6x + 33)(x+3) - 89}(x-1) + 46$$

$$= \underbrace{(3x^2 - 6x + 33)}_{q(x)} (x+3)(x-1) - \underbrace{89(x-1) + 46}_{r(x)}$$

So, quotient : $3x^2 - 6x + 33$

remainder : $-89(x-1) + 46 = -89x + 135$

Factor Theorem

A polynomial $d(x)$ is called a **factor** of another polynomial $f(x)$ if there exists a polynomial $h(x)$ such that

$$6 = 2 \cdot 3$$

$$f(x) = d(x)h(x)$$

If $d(x)$ is a factor of $f(x)$, we say also that $f(x)$ is **divisible** by $d(x)$ or that $f(x)$ is a **multiple** of $d(x)$.

$d(x)$ is a factor of $f(x)$ if and only if the remainder is zero when $f(x)$ is divided by $d(x)$.

Factor Theorem

Factor Theorem:

The polynomial $x - c$ is a factor of $f(x)$ if and only if $f(c) = 0$.

Proof:

The polynomial $x - c$ is a factor of $f(x)$
if and only if the remainder in the division
 $f(x)$ by $x - c$ is zero.

Since by the Remainder Theorem, the remainder
is equal to $f(c)$,

So it follows that $x - c$ is a factor of $f(x)$
if and only if $f(c) = 0$.

Factor Theorem

Example:

Determine whether or not $x - a$ is a factor of $f(x) = x^3 - 2x^2 + x - 2$ when $a = 1, i$, or $-i$.

$$\text{For } a=1, f(1) = (1)^3 - 2(1)^2 + (1) - 2 = -2 \neq 0$$

So $x-1$ is not a factor of $f(x)$

$$\begin{aligned}\text{For } a=i \quad f(i) &= i^3 - 2i^2 + i - 2 \\ &= -i + 2 + i - 2 = 0\end{aligned}$$

So, $x-i$ is a factor of $f(x)$.

$$\begin{aligned}\text{For } a=-i, f(-i) &= (-i)^3 - 2(-i)^2 + (-i) - 2 \\ &= i + 2 - i - 2 = 0\end{aligned}$$

So, $x+i$ is a factor of $f(x)$.

Factor Theorem

If $P(x)$ is a polynomial and $P(a) = 0$, where $a \in \mathbb{C}$, then a is called a **zero** of $P(x)$ or a **root** of the equation $P(x) = 0$

The following statements are equivalent:

- 1 $P(a) = 0$;
- 2 $x - a$ is a factor of $P(x)$;
- 3 a is a zero of $P(x)$;
- 4 a is a root of the equation $P(x) = 0$.

Example:

Let $P(x) = x^3 - 2x^2 + x - 2 = (x - i)(x + i)(x - 2)$.

- $P(i) = P(-i) = P(2) = 0$

- $x - i, x + i, x - 2$ are the factors of $P(x)$.

- $i, -i, 2$ are zeros of $P(x)$.

- $i, -i, 2$ are roots of $P(x) = 0$.